INCH-POUND

MIL-F-63518A (AR)
AMENDMENT 6
01 October 1992
SUPERSEDING
AMENDMENT 5
30 NOVEMBER 1989

MILITARY SPECIFICATION

FUZE, POINT DETONATING, M739A1
LESS BOOSTER PELLET

This amendment forms a part of Military Specification MIL-F-63518A (AR), dated 6 October 1986, and is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* Under Standards, Military, add:

MIL-STD-109 - Quality Assurance Terms and Definitions

PAGE 4

* Change paragraph 4.1 to read as follows:

4.1 Responsibility for inspection and standard quality assurance provisions: Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48708 shall apply and are hereby made a part of this detail specification.

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PAGE 5

* 4.3.1.f: Change "Fifty (50)" to "Forty-eight (48)".

PAGE 15

4.4.2.10: Delete minor 201 in its entirety.

PAGE 35

4.4.2.25: Change "9256645" to "9258646".

PAGE 36

4.4.2.25: Change "9256645" to "9258646".

PAGE 37

4.4.2.25: Change "9256646" to "9258646".

PAGE 49

4.4.2.34: Change "9256638" to "9258638".

PAGE 64

4.4.2.45: Change "9253634" to "9258634".

PAGE 73

4.4.2.52: For Critical 1, change "GAGE/VISUAL" to "GAGE"

4.4.2.52: Change "Gage" to "Gage, 4.5.32".

PAGE 94

4.4.2.72: Delete Major 105 in its entirety.

4.4.2.72: Add the following Major defect:

"108 Depth of detent retainer counterbore, min.; (a)., 3.1., Gage"

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* "4.5.13 Arming Test. Delete all subparagraphs and add the following:

4.5.13.1 Arming. The unarmed safing and arming module subassembly shall be subjected to this test by rotation approved automatic non-operator-dependent equipment. The safing and arming module subassembly shall be assembled to the applicable rotation holding fixture and rotated about its central axis in a counterclockwise direction when viewed from the top at 1700 + 100 revolutions per minute (R.P.M.). Any assembly that fails to arm shall be automatically rejected.

4.5.13.2 Arming time. The unarmed safing and arming module subassembly while being tested as specified in 4.5.13.1 shall be automatically tested for arming time. The time required for each module to reach the armed position shall be automatically determined and recorded by applicable test equipment. The acceptance or rejection shall be automatic.

The rotor shall be restrained from movement until the required R.P.M. is achieved and the arming time count shall begin automatically with the removal of the restraint. Any unit that arms in less than 750 milliseconds or more than 1200 milliseconds shall be automatically rejected. SPC monitoring of this test is mandatory and shall assure, as a minimum, that the product meets the arming time requirements, the process is in a continuous state of control, and that the Cpk value is not less than 1.33.

4.5.13.3 Locking. The armed safing and arming module subassembly subjected to the arming test in 4.5.13.1 shall be examined for the rotor being locked by the rotor lock pin. Any assembly that is not in the locked position after the arming test shall be rejected.

4.5.13.4 Qualification test. The Acceptance Inspection Equipment must pass a dynamic qualification test to demonstrate its reliability prior to its introduction into the production process for each contract. One thousand S&A assemblies shall be used for this test. One thousand consecutive correct decisions are required for the equipment to successfully pass the qualification test.

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It is in the contractor's interest to prepare assemblies as close to the product time limit requirements as practicable. The tolerance of the test samples is optional for the contractor provided that the 750 millisecond minimum time tolerance is in the plus direction (increasing time) and the 1200 millisecond (maximum time) is in the minus direction (for example: 750 + 50 - 0 and 1200 + 0 - 50). 500 of each standard shall be used in a process capability study to determine the machine capability at the product limits. Once established the proven test limits are to be considered the machine not-go reject limits for production and SPC.

4.5.13.5 Calibration and verification standards. The calibration and verification standards to be used in production shall be selected by the contractor and approved by the government. The time limits shall be within the product drawing limits and within the proven capability of the machine during the qualification test."

PAGE 106

4.5.27.1: for the 105MM M102, Zone 1, Delay Setting, change "8-in. Plywood" to "4-in. Plywood".

PAGE 108

4.5.27.2: for the 105mm M102, Zone 1, Delay Setting change "8-in. Plywood" to "4-in. Plywood".

PAGE 110

Add new paragraph 4.5.32 as follows:

"4.5.32 Gage for S&A module not in unarmed position. The gage must be capable of detecting the following conditions:

a. The rotor is in the safe position.

b. The setback pin and spring are present.

c. The setback pin is engaged in and locking the rotor.

d. Both spinlocks are present and engaged in the rotor.

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Of primary importance is that the gage, whether semi-automatic or fully automatic, must make the decision as to the safe position of the rotor in the S&A. The decision for safety of the rotor is not to be at the discretion of the operator/inspector.

The use of or operation of the gage should be such that the rejected S&A is controlled so that it cannot be inadvertently added to the accepted product. A rigid procedure for materiel handling of critical defects must be part of the operation/use of any gage inspection for such a defect.

The inspection must be the last operation prior to assembling the S&A/retainer assembly into the body of the fuze. This means there is a smooth, continuous flow of accepted product between the gaging station and the assembly operation. It is desirable that accepted units be automatically marked for identification.

The gage must pass a qualification test to demonstrate its reliability prior to its introduction into the production process. 1000 S&A/retainer assemblies shall be used for this purpose and they shall be of the following description:

300 properly assembled, fully safe

175 with both spinlocks missing (independent detection of each)

175 with setback pin missing

175 with setback pin and both spinlocks missing

175 with rotor, setback pin, and both spinlocks missing

When tested with all 1000 of the above masters, the gage must pass the test with no incorrect decisions.

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* Paragraph 6.2 add the following paragraph:

"6.2.1 Contract data requirements. When this specification is used in acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs."

Ref. Para.	DID Number	DID Title	Suggested Tailoring
3.4	DI-R-5297	Quality Program Plan	None

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6.8: Add the following at the end of the paragraph:

"Fuzes set delay which fail to function as intended behind the target, but which do function on subsequent impact (FSI), shall not be classed as duds and shall be reported 'for information only'."

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* Add the following paragraph:

"6.11 Submission of statistical process control plan. The contractor will submit a SPC plan to the contracting officer with copies forwarded to ARDEC, Picatinny Arsenal, New Jersey 07806-5000, ATTN: AMSMC-QAR-I(D) and SMCAR-AEF-F as specified in DD form 1423.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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